ML 1.2

Saturday, February 17, 2024 9:06 PM

Intro to ML and NN

1.0

- Linear Regression
- Logistic Regression

Evaluate

- Train
- Validation
- Test

- overfit
- underfit

→ Regularization

- L_1
- L_2

Linear Regression

- RSS
- R^2
- MSE

Logistic Regression

↓ classification

- Accuracy

Imbalanced

- Precision — FP ↓
- Recall — FN ↓
- F1 score

Threshold ⇐ ROC AUC TPR / FPR

Imbalanced

↓

P R curve

Precision Recall

Imbalance

under over SMOTE

Confusion — { 0 , 1

{PCA} 7 ~~3~~ 2 Visual

* yes — Logistic but its is of

No use

↓

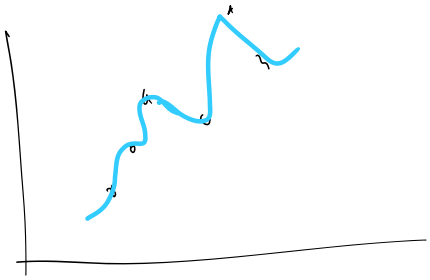
Log loss = { $w^T x + w_0$

Linearly separeal

Linearly Separable

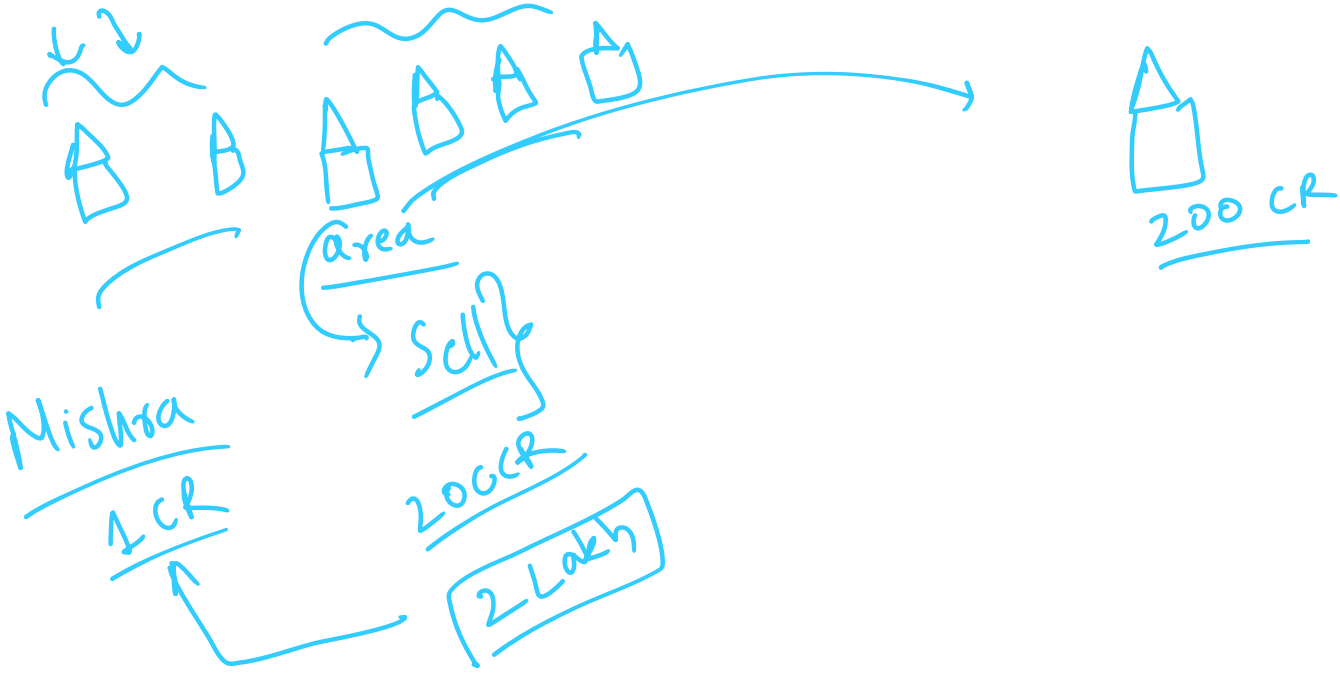
Polynomial → overfit

Multiclass → OVR

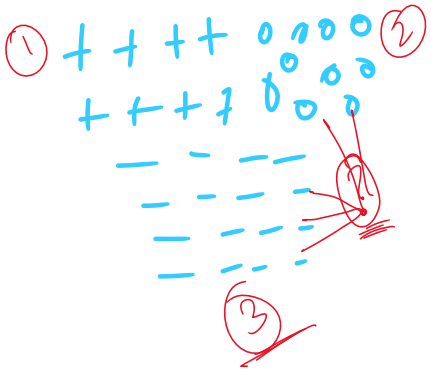


overfit

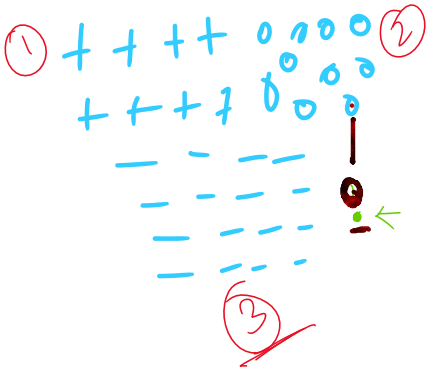
Linear



K Nearest Neighbor

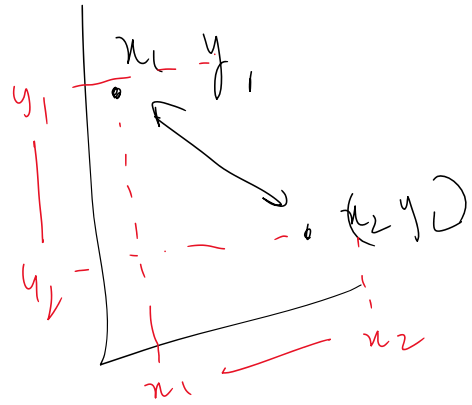


f_1 Marks Eng	f_2 Math	f_3 Bio	Class
92	93	94	Topper
91	94.5	96	Topper
60	30	40	Class 2



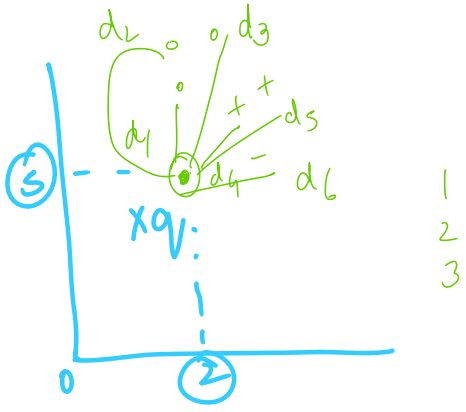


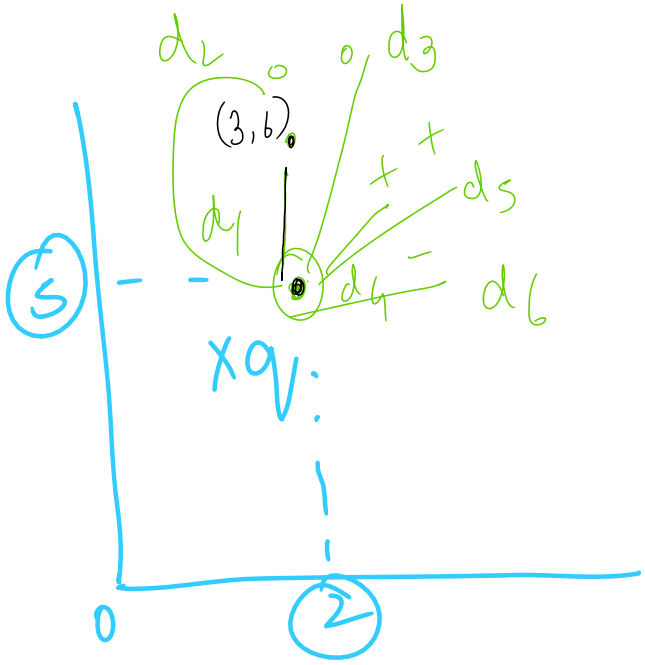
distance



$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$





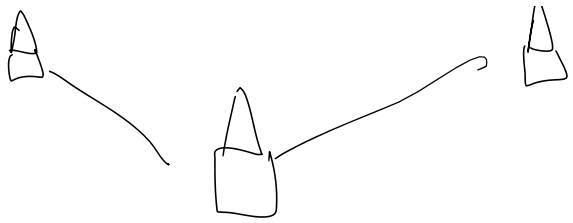


$$\sqrt{(2-3)^2 + (5-6)^2}$$

$$\sqrt{1^2 + 1^2}$$

$$\sqrt{2}$$

$$1.41$$



Nearest — Shortest distance

{ 3, 5, ≠, 9 } Hyperparameter

K

K — N Nearest N Neighbor

→ top 3
→ top 5

Logistic

Weights

$\{w_1 \quad w_2 \quad w_0\}$

Learn

Parametric

KNN

No weights

d_1

Take the Shortest

Parametric

Non Parametric

the shortest

1

d_1 — 1

s1- find majority vote
s2- perform euclidean distance
s3- sort and select k datapoints
s4- give class to x_q datapoint

From <https://www.scaler.com/instructor/meetings/i/ml-k-nearest-neighbors-1-9/>

S_2 — dist

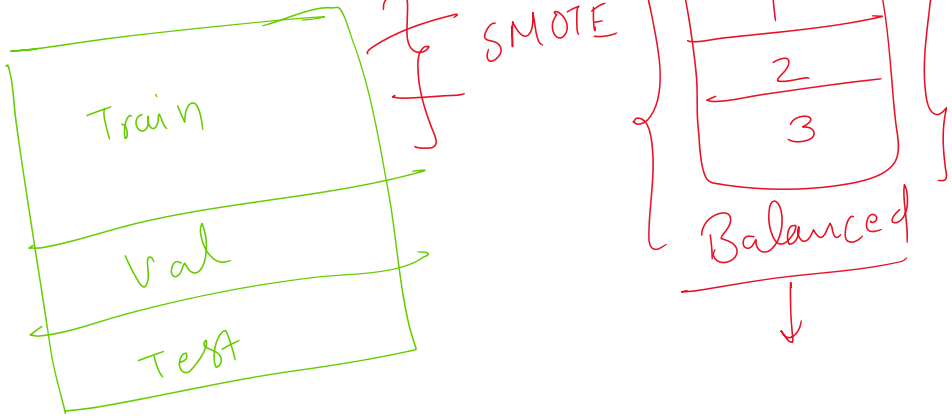
S_3 — 3 Top

S_1 — vote

S_4 →

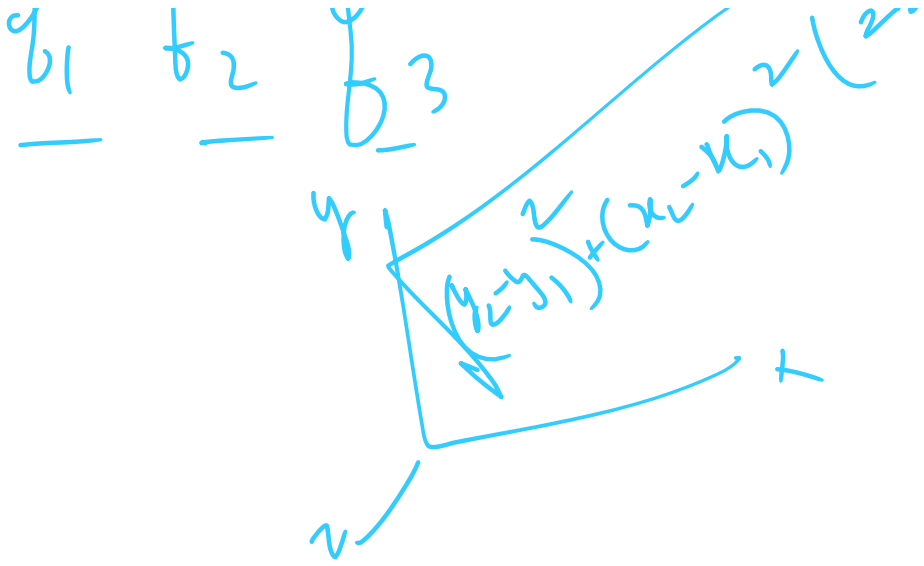
Algo — understand
Math
code

} Limitation
Assumption



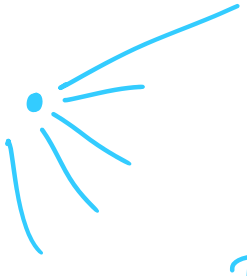
f_1 f_2 f_3

$2^n(2^n-2)^k$



Test — 110 Datapoint

Train + val
300 Points



Distance →

NN — RNN → Transformers → LLM